ZLAC706-CAN

Low Voltage Servo Driver Manual

(Special for Hub Motor)

[Please read the manual carefully before use to avoid damage to the driver]

Version Update Instructions

Create/modify Date	Modifier	Version NO.	Remark
2017-07-28	PC	V1.0	
2017-08-19	PC	V1.1	
2017-12-12	PC	V1.2	Increase the function to change current and torque under RS232, CAN communication and PID parameter address setting.
2019-4-16	PC	V1.3	Increase read alarm status, voltage, current, speed, position reference, position feedback separately under RS232, CAN communication. Increase CAN communication emergency stop command (driver version 18206)

CATALOGUE

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PREFACE

Thanks for choosing the easy-use low voltage servo driver.

This manual describes the installation, debugging, maintenance, operation and other aspects of this servo driver(25W-400W). Before using, please read the manual and make clear of the safety precautions of this product.

This manual will be timely revised when product is improved. It will not be notified when specification and version are changed for some other reasons.

Any questions when using our products, please read the relevant manual or call our technical service department, we will meet your requirements in the shortest possible time.

Marks and warning signal:

Danger: Indicates that this operation error may endanger personal safety!

Attention: Indicates that this operation error may result in equipment damage!

SAFETY PRECAUTIONS

Open package and Check

Don’t install controllers which is missing parts or damaged.

The servo driver must be used with a matching servo motor.

Installation

Please installed the driver on a non-flammable metal frame, to prevent the intrusion of dust, corrosive gases, conductive objects, liquids and flammable materials, and maintain good heat dissipation conditions.

When installing, be sure to tighten the mounting screws of the driver. Please prevent that servo driver and servo motor from vibration and withstand shock.

Wiring

Please make wiring work by professional electrical engineer.

Before wiring, please confirm that the input power is off. Wiring and inspection must be performed after the power is turned off and the driver indicator is off to prevent electric shock.

Make sure that the driver indicator is off before plugging and unplugging the driver terminals.

The grounding terminal PE should be grounded reliably through the left screw of the driver.

Please set the emergency stop circuit outside the controller.

Do not connect the power input cable to output U, V, W terminals.

Please tighten the output terminal with a suitable torque.

Electrify

Please confirm whether the main circuit input power is consistent with the rated working voltage of the driver.

Do not make high voltage and insulation resistance test for driver for high voltage and insulation resistance at will.

Do not connect the electromagnetic contactor or electromagnetic switch to the output circuit.

Operation

Do not touch the output terminals directly after the driver is powered on.

When the system is on operation, the temperature of driver and motor may rise, do not touch it.

Please confirm the input and output signals to ensure safe operation.

The alarm can be reset only after the operation signal is cut off. Alarm resetting in the running signal state will cause the driver to restart suddenly.

Do not change the parameter settings of the driver at will. The parameter modification needs to be performed under standby condition.

Maintenance and Inspection

Do not touch the controller terminals directly, as some of them have high voltage, and it's very dangerous.

Before powering up, be sure to install the cover. When removing the cover, be sure to cut off the power supply first.

Before wiring, please confirm whether the input power is off.

After cutting off the main circuit input power and confirming the driver indicator light has completely extinguished, it can be inspected and maintained.

Do the inspection and maintenance by professional electrical engineer.

Do not do wiring, disassembling or other operation on the terminals during power on.

There is an integrated circuit on the main control board of the driver. Please pay full attention when checking to avoid damage caused by static induction.

OUTLINE

The low-voltage servo driver is developed with a high-performance processor to provide users with a cost-effective servo control solution.

Under the premise of ensuring stability and reliability, we pursue the functions and performance that are closest to the application. Compared with stepping products, it has low noise, low heat generation, high speed, constant torque output, and no lost step. Compared with step servo products, it completely eliminates the inherent disadvantages of stepping products, and has better functions, performance and reliability. Compared with well-known foreign high voltage servo, the performance is close, price is lower, and it is easy to use.

1. BASIC FEATURES

Working voltage: 24V-60VDC (24V-36VDC for 200W motor, 24V-60VDC for 400W motor).
Output current: average 8A, peak 24A.
Speed: up to 1000RPM for hub motor.
Adapter motor: hub motor within 400W.
Control mode: external pulse, analog quantity, host computer communication control etc., support position, speed and torque mode.
Parameter commissioning: RS232 communication, PC debugging software write once, can backup and import parameters.
Communication method: CAN, RS232.
Abnormal protection: has functions of under-voltage, over-voltage, overload, over-current, encoder abnormality, etc. and has alarm output.

2. APPLICATION

Various types of electronic processing equipment, assembly line material conveying device, medical equipment, instrumentation, precision testing equipment, channel gate control, cartesian coordinates robot, servo fixed length positioning, garage blocking control, equipment loading and unloading device, equipment auxiliary exercise device, grab and handling machinery, inkjet printers, photo machines, home and office automation devices etc.

3. TECHNICAL NORMS

Adopt FOC magnetic field orientation control technology and SVPWM space vector modulation algorithm can modify the motor’s parameters easily to adapt to various motors with different specifications. Built-in electronic gear, graphical debugging and monitoring software. The control function can be customized according to the user's needs to integrate control and drive.

Repeat tracking error: 1Pulse.
Speed control accuracy: 2RPM.
Receiving frequency: 600KHZ.
Maximum speed: 1000RPM for hub motor.
Minimum speed: 1RPM.
Positioning accuracy: 1/10000.
Highest no-load acceleration: 200RPM/MS.
Adaptable to hub motor within 24V/36V/48V/60V 400W.

4. INTERFACE DEFINITION

4.1 Power/Motor Interface(P3)

No.	Mark	Name	Remark
1	GND	Input power-	DC 24V-60V
2	VDC	Input power+	
3	U	Motor power line U phase	Must be connected to the motor one by one according to the mark
4	V	Motor power line V phase	
5	W	Motor power line W phase	

Tab 4-1

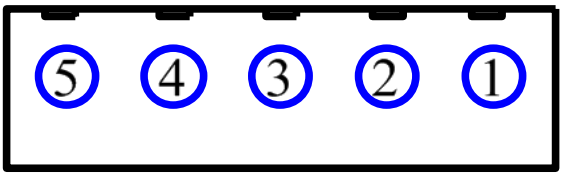


Fig 4-1 Power/motor interface terminal diagram

4.2 Encoder Interface(P2)

No.	Mark	Name	No.	Mark	Name
1	GND	Output power ground	7	B+	Encoder B phase positive input
2	VCC	Output power+5V	8	A+	Encoder A phase positive input
3	W+	Encoder W phase positive input	9～12		No connection
4	V+	Encoder V phase positive input	13	Z-	Encoder Z phase negative input
5	U+	Encoder U phase positive input	14	B-	Encoder B phase negative input
6	Z+	Encoder Z phase positive input	15	A-	Encoder A phase negative input

Tab 4-2

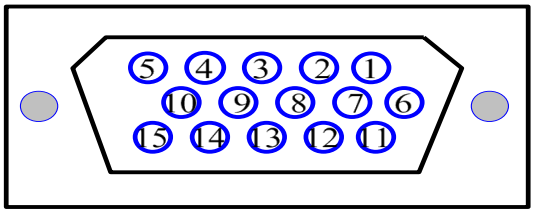


Fig 4-2 Encoder interface terminal diagram

4.3 Control Interface(p1)

No.	Mark	Name	Remark
1	ERC-	Enable negative input	Enable signal: This signal is used to enable or disable the servo motor. When the ERC+ and ERC- are at high level, the drive will cut off the motor power, so that the motor is in a free state and does not respond to the pulse. When the ERC+ and ERC- are low, the motor is enabled. If use 12V or 24V, need to string 1.5-2.2K resistor.
2	ERC+	Enable positive input	
3	DIR-	Direction negative input	Direction signal: When the input between DIR+ and DIR- is at high level , it is reversed, and vice versa. The DIR signal must be established at least 5uS ahead of PUL signal, 4-5V at high level and 0-0.5V at low level. If use 12V or 24V, need to string 1.5-2.2K resistor.
4	DIR+	Direction positive input	
5	PUL-	Pulse negative input	Pulse signal: Pulse rising edge is effective, 4-5v at high level, 0-0.5v at low level, the pulse width should be greater than 1.6uS. If 12V or 24V is used, need to string 1.5-2.2k resistor.
6	PUL+	Pulse positive input	

Tab 4-3

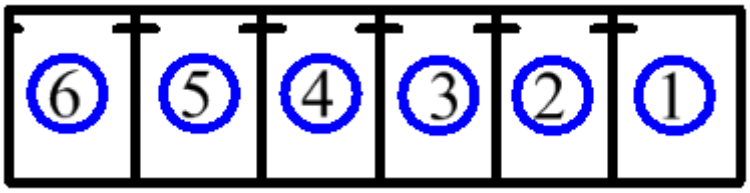


Fig 4-3 Control interface terminal diagram

4.3.1 Control signal input example

Control signal differential mode input, control signal interface wiring diagram, as shown in figure 4-4.

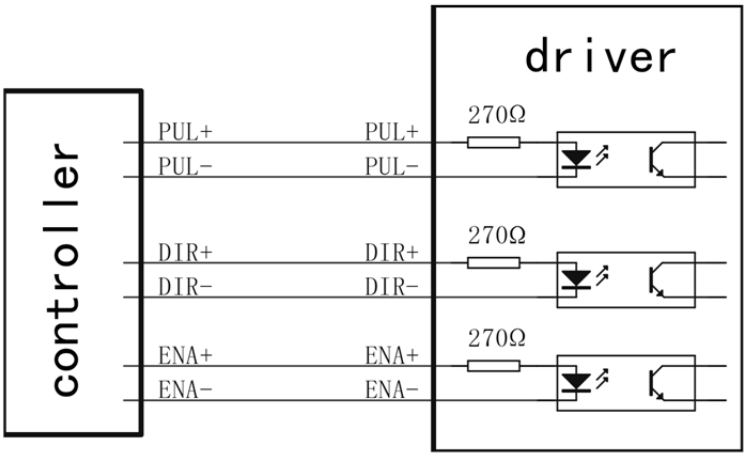


Fig 4-4 Wiring diagram of differential mode control signal interface

The wiring diagram of the single-ended mode control signal interface as figure 4-5.

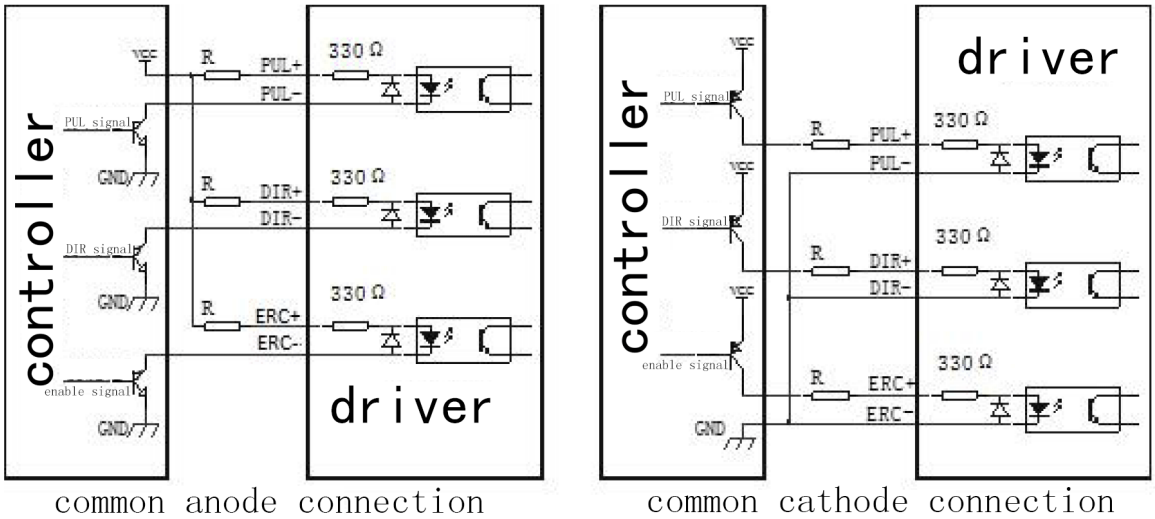


Fig 4-5 Single-ended mode control signal interface wiring diagram

Special reminder: when the control signal is 12V or 24V, you need to connect an external current limiting resistor. Connect 1K 1/8W resistor for 12V pulse, 2K 1/8W resistor for 24V pulse. Otherwise, long-term operation will damage the opto-isolated device of driver.

4.4 OUT/IN Interface

Terminal No.	Name	Remark	Terminal No.	Name	Remark
1	5V	Output +5V power	2	GND	Power ground
3	EA+	Encoder A phase positive output	4	EA-	Encoder A phase negative output
5	EB+	Encoder B phase positive output	6	EB-	Encoder B phase negative output
7	EZ+	Encoder Z phase positive output	8	EZ-	Encoder Z phase negative output
9	AI	Analog input	10	GND	Power ground
11	FAULT	Alarm signal output	12	COM	Output power ground

Tab 4-6

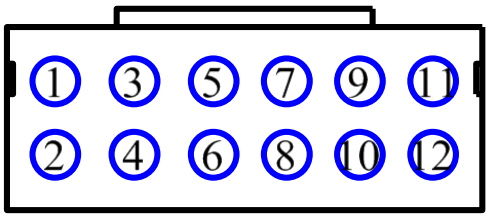


Fig 4-6

4.4.1 Analog input control example

The analog external potentiometer is shown in figure 4-7. There are two ways to control the analog control, which can be selected by RS232 debugging software: position control mode - external analog input and speed control mode - external analog input.

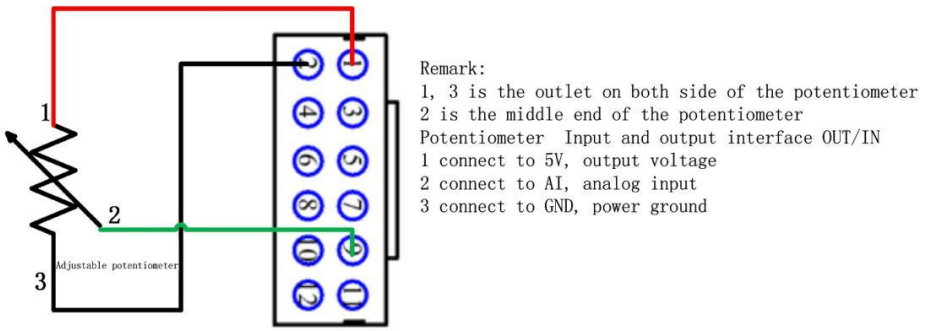
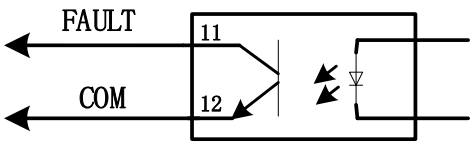


Fig 4-7 Analog quantity external potentiometer diagram

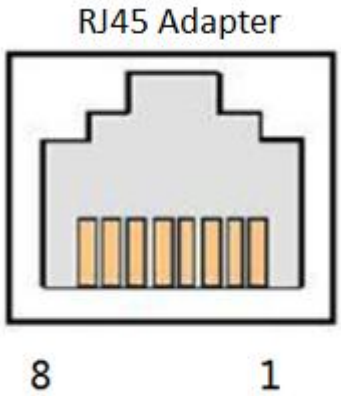
4.4.2 Alarm signal output example

The FAULT port is an opto-isolated OC output with the maximum withstand voltage 30VDC and maximum saturation current 50mA. When the driver is working normally, the output optocoupler is not conductiv



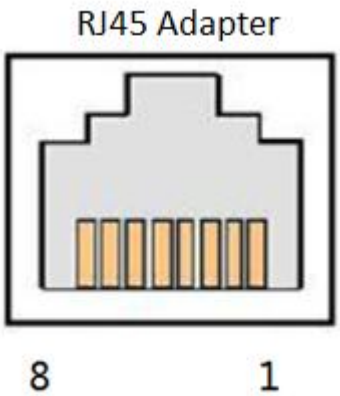
4.5 Communication Interface

Interface RS232/CANIN

Diagram	Terminal No.	Name	Remark
	1	CANH	CANH high signal
	2	CANL	CANL low signal
	3	GND	Signal ground
	4	RX	RS232 signal received
	5～7	NC	No connection
	8	TX	RS232 signal transmitted

Tab 4-7

Communication Interface CANOUT

Diagram	Terminal No.	Name	Remark
	1	CANH	CANH high signal
	2	CANL	CANL low signal
	3	GND	Signal ground
	4～8	NC	No connect

Tab 4-8

5. CONTROL MODE

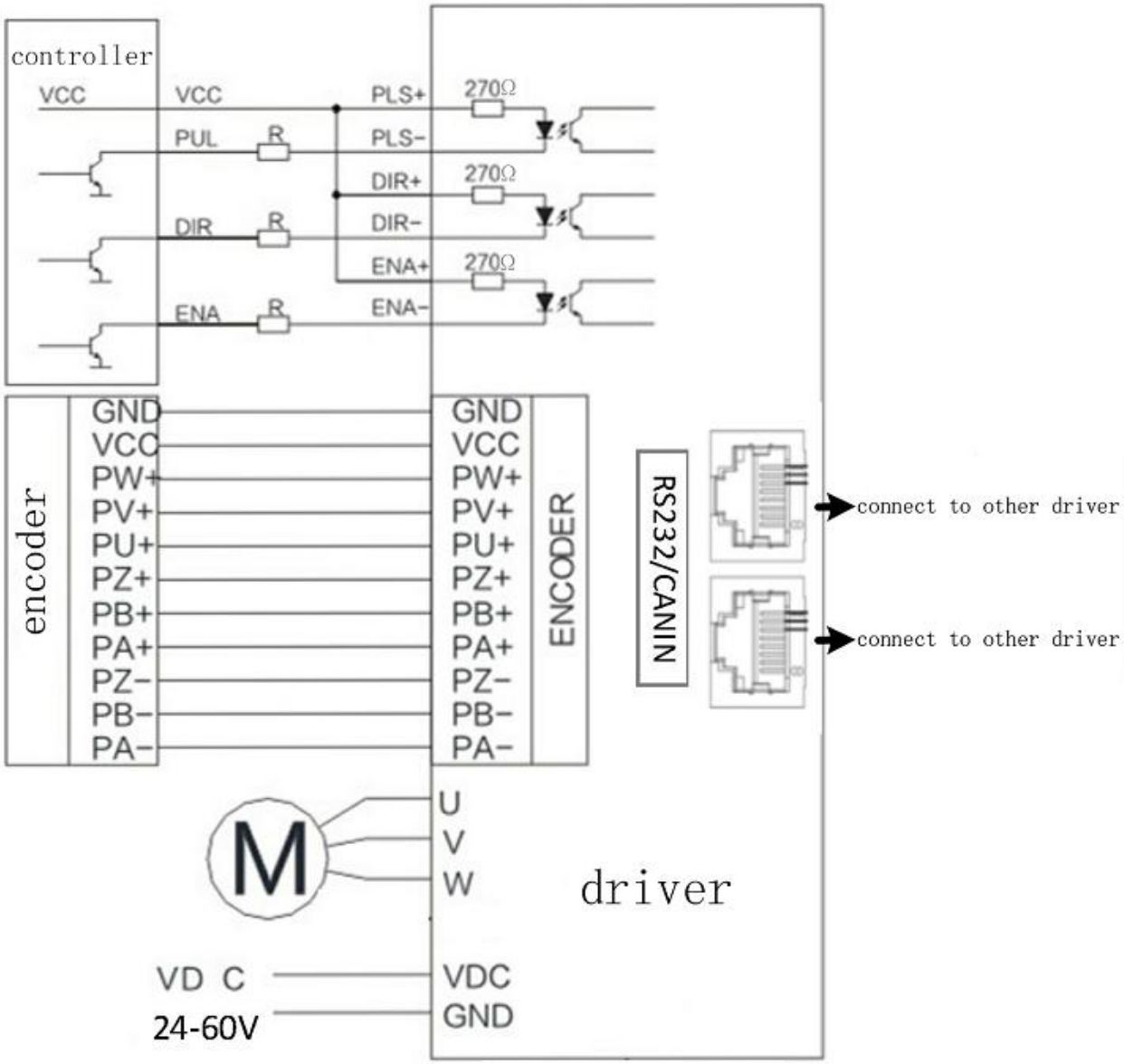
5.1 Basic Control

The driver provides 3 basic operation modes: position, speed and torque, using a single control mode. All operation modes and instructions are listed below.

Control mode selection	Control source selection	Instruction
Position control mode	External pulse input	The driver accepts a position command to control the motor to the target position. The position command is input by the terminal, and the signal type is pulse + direction
	PC digital input	Relative position: The driver enable start time is mechanical 0 point, each time the PC digital input value is written, the motor rotates the target distance. Absolute position: The driver enable start time is mechanical 0 point, each time the PC digital input value is written, the motor rotates to the target position with the mechanical 0 point as the reference.
	External analog input	When the external analog input is 0~+5V, the motor rotates -16384~+16384 pulses in absolute position mode
Speed control mode	PC digital input	Servo motor input range: -4000RPM~+4000RPM; Hub motor input range: -1000RPM~+1000RPM
	External analog input	When the external analog input is 0~+5V, the servo motor runs at a speed of 4000RPM~+4000RPM; Hub motor at a speed of - 1000RPM ~ +1000RPM
Torque control mode	PC digital input	The input range is -7500~+7500. The corresponding output current is 24A.When the input is positive, it corresponds to the positive torque, and the negative value is the reverse torque.
	External analog input	When the external analog input is 0~+5V, the corresponding digital torque range is: -16384~+16384. Drive internal limiting to a maximum of 7500.

Tab 5-1

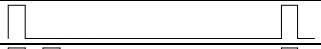
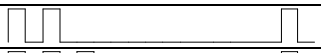
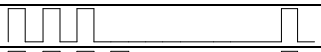
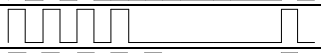
6. TYPICAL WIRING



Note: When VCC is 5V, R short-circuited;
When VCC is 12V, R is 1K, greater than 0.125W resistor.
When VCC is 24V, R is 2K, greater than 0.125W resistor;

7. INDICATOR LIGHT AND FAULT DIAGNOSIS

Green LED is the power indicator, which is always on when the driver is powered up; it is off when the driver is powered off.
Red LED is the fault indicator. When a fault occurs, the LED flashes in a cycle of 5 seconds. When the fault is cleared, it keeps off. It flashes at 2Hz, with the LED on for 200ms and off for 300ms. Its flash number in 5 seconds represents different fault information, as shown in the following table:

NO.	Flash waveform	Red indicator flash waveform	Fault description
1	1		Overload fault
2	2		Under-voltage fault(VDC≤16V)
3	3		Over-voltage fault(VDC≥80V)
4	4		Encoder falling fault
5	5		Over-current fault

Users can also read the corresponding fault code through the PC host computer and serial text.

8. COMMUNICATION PROTOCOL

8.1 CAN Bus Protocol

8.1.1 CAN bus protocol

ZL-C communication speed is 500K, the protocol is a signed integer data with 16-bit length or 32-bit length which is from data of each instruction like ID, slave number, function code, register data address, register data subaddress, data content 1 high 8 bits, data content 1 low 8 bits, data content 2 high 8 bits, data content 2 low 8 bits, and they form a complete CAN communication command.
The specific format is as follow:

(1) Write data, and the data will not be saved when power-off. The host computer sends protocol data command, after the slave receives correct command, it will return the corresponding data command.
The command format is as follow:

Slave ID	Data Field							
Preset	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
	slave number	function code	register address	register subaddress	high 8 bits	low 8 bits	high 8 bits	low 8 bits
	preset	0xDA	0x00	(see parameter mapping table for details)	——	——	——	——

If the received data is correct, the function code will change to 0xFB, other messages will return in original text. Format is as follow:

Slave ID	Data Field							
Preset	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
	slave number	function code	register address	register subaddress	high 8 bits	low 8 bits	low 8 bits	low 8 bits
	preset	0xDB	0x00	(see parameter mapping table for details)	——	——	——	——

If the address does not exist, the function code will change to 0xFF, and other message will return in original text. Format is as follow:

Slave ID	Data Field							
Preset	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
	slave number	function code	register address	register subaddress	high 8 bits	low 8 bits	high 8 bits	low 8 bits
	preset	0xFF	0x00	(see parameter mapping table for details)	——	——	——	——

(2) Read data. The host computer sends the protocol data command, after the slave receives correct command, it will return the corresponding data command.

The command format is as follow:

Slave ID	Data Field							
Preset	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
	slave number	function code	register address	register subaddress	high 8 bits	low 8 bits	high 8 bits	low 8 bits
	Preset	0xDC	0x00	(See parameter mapping table for details)	0x00	0x00	0x00	0x00

If the received data is correct, the function code will change to 0xDB, other messages will return in original text. Format as follows:

Slave ID	Data Field							
preset	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
	slave number	function code	register address	register subaddress	high 8 bits	low 8 bits	high 8 bits	low 8 bits
	preset	0xDB	0x00	(See parameter mapping table for details)	——	——	——	——

If the address does not exist, the function code will change to 0xDF, and other message will return in original text. Format is as follow:

Slave ID	Data Field							
preset	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
	slave number	function code	register address	register subaddress	high 8 bits	low 8 bits	high 8 bits	low 8 bits
	preset	0xFF	0x00	(see parameter mapping table for details)	——	——	——	——

(3)Position mode related parameter mapping list:

Position Control Mode				
Internal Add.	**Variable Name**	**Set Value**	**Message(ID=1)**	**Remark**
0x0010	operation control word	F	Send 00 DA 00 10 00 00 00 0F Return 00 DB 00 10 00 00 00 0F	release motor (downtime)
0x0010	operation control word	1F	Send 00 DA 00 10 00 00 00 1F Return 00 DB 00 10 00 00 00 1F	enable motor(start)
0x0019	Working mode	3F	Send 00 DA 00 19 00 00 00 3F Return 00 DB 00 19 00 00 00 3F	position mode
0x0017	position mode control word	4F	Send 00 DA 00 17 00 00 00 4F Return 00 DB 00 17 00 00 00 4F	absolute position operation
0x0017	position mode control word	5F	Send 00 DA 00 17 00 00 00 5F Return 00 DB 00 17 00 00 00 5F	relative position operation
0x0018	origin control word	6F	Send 00 DA 00 18 00 00 00 6F Return 00 DB 00 18 00 00 00 6F	run to Z signal (hub motor doesn't have Z signal)
0x0012	position mode time of ACC/DEC	10*100 MS	Send 00 DA 00 12 00 00 0A 0A Return 00 DB 00 12 00 00 0A 0A	ACC time from 0 to the highest speed
0x0014	trapezoid speed	1000RPM (0x1000)	Send 00 DA 00 14 00 00 0A AA Return 00 DB 00 14 00 00 0A AA	Max. speed or uniform speed
0x0015	alarm reset	7F	Send 00 DA 00 15 00 00 00 7F Return 00 DB 00 15 00 00 00 7F	clear when driving alarm
0x0016	target position	123456 (0x1E240)	Send 00 DA 00 16 00 01 E2 40 Return 00 DB 00 16 00 01 E2 40 high 16 bits: 00 01 low 16 bits: E2 40	0x16 address stores 32-bit position command: high 16 bits and low 16 bits
0x0030	Software brake	1F	Send 00 DA 00 30 00 00 00 1F Return 00 DB 00 30 00 00 00 1F	Emergency stop

Note: In position mode, start the motor before sending the target position command. If the motor is not started, the command is invalid. Therefore, the target position command must be set after all the basic settings are completed.

(4) Speed mode related parameter mapping list:

Speed Control Mode				
Internal Add.	**Variable Name**	**Set Value**	**Message(ID=1)**	**Remark**
0x0010	operation control word	F	Send 00 DA 00 10 00 00 00 0F Return 00 DB 00 10 00 00 00 0F	release motor (downtime)
0x0010	operation control word	1F	Send 00 DA 00 10 00 00 00 1F Return 00 DB 00 10 00 00 00 1F	enable motor (start)
0x0011	Target speed	500RPM (0x1000)	Send 00 DA 00 11 00 00 05 55 Return 00 DB 00 11 00 00 05 55	motor speed
0x0013	speed mode time of ACC/DEC	10*100MS	Send 00 DA 00 13 00 00 0A 0A Return 00 DB 00 13 00 00 0A 0A	ACC time from 0 to the highest speed
0x0015	alarm reset	7F	Send 00 DA 00 15 00 00 00 7F Return 00 DB 00 15 00 00 00 7F	clear when driving alarm
0x0019	working mode	2F	Send 00 DA 00 19 00 00 00 2F Return 00 DB 00 19 00 00 00 2F	speed mode
0x0030	Software brake	1F	Send 00 DA 00 30 00 00 00 1F Return 00 DB 00 30 00 00 00 1F	Emergency stop

PID Parameter mapping list				
Internal Add.	**Variable Name**	**Set Value**	**Message(ID=1)**	**Remark**
0x0020	position loop Kp	3000 (0x0BB8)	Send 00 DA 00 20 00 00 0B B8 Return 00 DB 00 20 00 00 0B B8	
0x0021	position loop Kd	1 (0x0001)	Send 00 DA 00 21 00 00 00 01 Return 00 DB 00 21 00 00 00 01	
0x0022	Position loop feed forward gain Ki	0 (0x0001)	Send 00 DA 00 22 00 00 00 00 Return 00 DB 00 22 00 00 00 00	Set the value to 0 or 1
0x0023	Speed loop proportional gain Kp	5000 (0x1388)	Send 00 DA 00 23 00 00 13 88 Return 00 DB 00 23 00 00 13 88	
0x0024	speed loop Ki	500 (0x01F4)	Send 00 DA 00 24 00 00 01 F4 Return 00 DB 00 24 00 00 01 F4	
0x0025	speed loop Kd	0 (0x0000)	Send 00 DA 00 25 00 00 00 00 Return 00 DB 00 25 00 00 00 00	

0x0026	current loop Kp	500 (0x01F4)	Send 00 DA 00 26 00 00 01 F4 Return 00 DB 00 26 00 00 01 F4	
0x0027	current loop Ki	500 (0x01F4)	Send 00 DA 00 27 00 00 01 F4 Return 00 DB 00 27 00 00 01 F4	

(5) PID related parameter mapping list:

Under the default parameter setting, if the motor vibrates, reduce the value of the speed loop Kp, and increase the value of the speed loop KI appropriately when vibrating stops. If the reduced speed loop Kp is not enough to vibrate the motor , please reduce the value of the position loop Kp. If the motor doesn't vibrate, please increase the value of the speed loop Kp slowly and reduce the value of the speed loop Ki. Under the condition of satisfying the rigidity, increasing the value of the speed loop Kd appropriately can reduce the vibration of the motor, and increasing Kp excessively will cause high-frequency noise, please pay attention to it. If the above steps cannot reduce the vibration, you could reducr value of the current loop Kp appropriately.

Internal Add.	Variable Name	Message(ID=1)	Remark
0x00E1	bus voltage	Send 00 DC 00 E1 00 00 00 00 Return 00 DB 00 E1 00 00 00 19	Voltage = 0x19 corresponds to 25V
0x00E2	output current	Send 00 DC 00 E2 00 00 00 00 Return 00 DB 00 E2 00 00 00 40	Current = 0x40 corresponds to 64mA
0x00E3	alarm status	Internal Control (PC): Send 00 DC 00 E3 00 00 00 00 Return 00 DB 00 E3 00 00 00 00 Return 00 DB 00 E3 00 00 00 01 Return 00 DB 00 E3 00 00 00 02 Return 00 DB 00 E3 00 00 00 04 Return 00 DB 00 E3 00 00 00 08 Return 00 DB 00 E3 00 00 00 10 Return 00 DB 00 E3 00 00 00 20 Return 00 DB 00 E3 00 00 00 40 External Control (PLC): Send 00 DC 00 E3 00 00 00 80 Return 00 DB 00 E3 00 00 00 80 Return 00 DB 00 E3 00 00 00 81 Return 00 DB 00 E3 00 00 00 82 Return 00 DB 00 E3 00 00 00 84 Return 00 DB 00 E3 00 00 00 88 Return 00 DB 00 E3 00 00 00 90 Return 00 DB 00 E3 00 00 00 A0 Return 00 DB 00 E3 00 00 00 C0	 Stop state Startup state Overcurrent Overvoltage Encoder error Overheat Undervoltage Overload Stop state Startup state Overcurrent Overvoltage Encoder error Overheat Undervoltage Overload
0x00E4	real-time	Send 00 DC 00 E4 00 00 00 00	Actual speed = set value /

	speed	Return 00 DB 00 E4 00 00 00 96	8192 * 3000 0x00 0x96 corresponds to 150 Speed = 150/8192*3000=54rpm
0x00E6	position given	Send 00 DC 00 E6 00 00 00 00 Return 00 DB 00 E6 00 00 A0 00	0xA0 0x00 corresponds to 40960
0x00E8	position feedback	Send 00 DC 00 E8 00 00 00 00 Return 00 DB 00 E8 00 00 35 CF	For reverse direction, value is negative, taking the complement of the target position (inverting +1) 0x35 0xCF corresponds to 13775

(6) readable parameter mapping list

(7) Other related parameter mapping list:

Internal Add.	Variable Name	Variable Name	Variable Name	Remark
0x002d	torque mode	8000 （0x1F40）	Send 00 DA 00 2d 00 00 1F 40 Return 00 DB 00 2d 00 00 1F 40	8000mA
0x0019	motion control word	8F	Send 00 DA 00 19 00 00 00 8F Return 00 DB 00 19 00 00 00 8F	torque mode
0x0008	target torque	1500 (0x05DC)	Send 00 DA 00 08 00 00 05 DC Return 00 DB 00 08 00 00 05 DC	2500 corresponds to 8A, 1500 corresponds to 4.8A

(8) CAN heartbeat message

In the PC software, you can set the upload interval of heartbeat message. If the time is set to 0, then stop the function of uploading heartbeat message. If it is greater than 0, then upload the selected content automatically.

The position command is the position pulse of the actual feedback. Current command is 100 times the actual output current. Speed command is a digital command, and the specific correspondence is: actual speed = digital command / 8192 * 3000RPM.

Heartbeat message NO.	Heartbeat message content	Message
0	position, current, speed	00 FE 00 20 00 00 00 00 00 FE 00 21 00 0A 00 00
1	position	00 FE 00 20 00 00 00 00
2	position	00 FE 00 21 00 07 00 00

Heartbeat message description:

00 FE 00 20 **00 01 E1 E2** 0X00 01 E1 E2 = 123362 is the position of feedback

00 FE 00 21 **00 1E 04 3F** 0X00 1E= 30 Current=30/100=0.3A

0X04 3F=1087 Speed=1087/8192*3000RPM=398RPM

If the driver alarms, the heartbeat message is forced to report the alarm code:

00 internal alarm code Add 00 to alarm code in internal control mode

80 external control code Add 80 to alarm code in external control mode

	Internal Control (PC)	External Control (PLC)
over-current	**00 FF 00 EB 02 02 02 02**	**00 FF 00 EB 82 82 82 82**
over-voltage	**00 FF 00 EB 04 04 04 04**	**00 FF 00 EB 84 84 84 84**
Encoder failure or position error is too large	**00 FF 00 EB 08 08 08 08**	**00 FF 00 EB 88 88 88 88**
overheat	**00 FF 00 EB 10 10 10 10**	**00 FF 00 EB 90 90 90 90**
under-voltage	**00 FF 00 EB 20 20 20 20**	**00 FF 00 EB A0 A0 A0 A0**
overload	**00 FF 00 EB 40 40 40 40**	**00 FF 00 EB C0 C0 C0 C0**

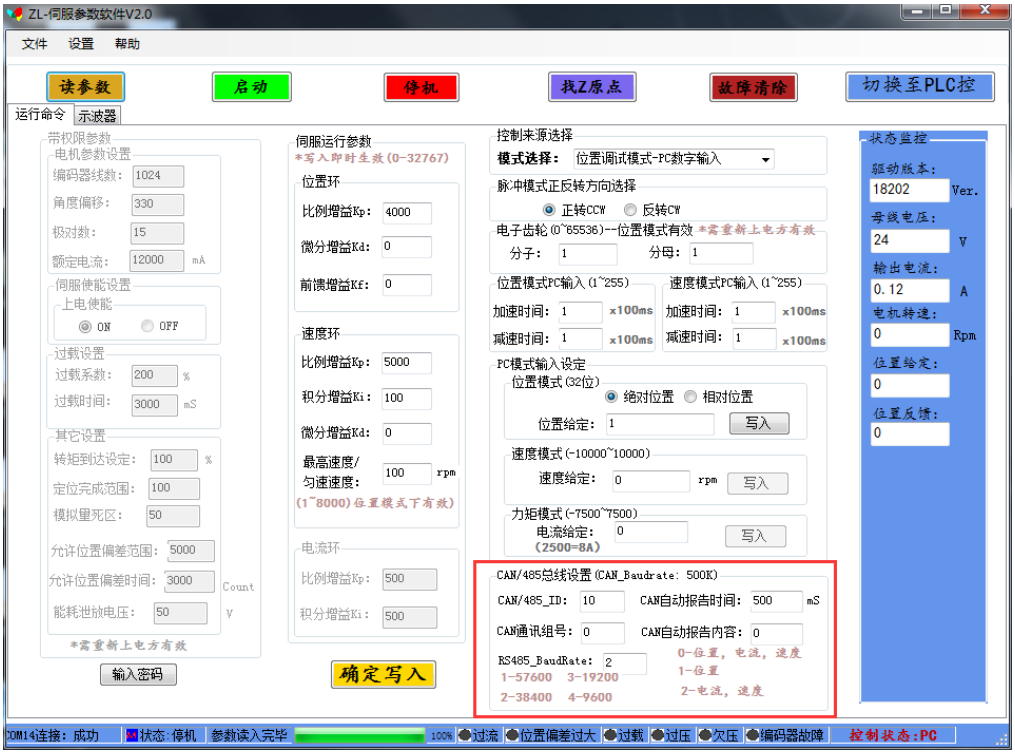
8.1.2 CAN communication application example

Driver internal speed conversion: actual speed = set value / 8192 * 3000RPM.

1. Servo USB to serial port host computer settings

First, connect the device and turn on the power.

The CAN bus baud rate defaults to 500Kbps. The baud rate can be modified in RS485_BaudRate: input 1 (represented as 1MHz baud rate), input 2 (500KHz baud rate), input 3 (250KHz baud rate), input 4 (125KHz Baud rate). After modifying the baud rate, please power off and restart the driver to finish the modification successfully. Set frame ID, automatic report time, control mode PC control as shown in the figure: CAN_ID is 10 (0-109), the automatic reporting time is 500ms (value according to the situation), and RS485_BaudRate is 2 to represent 500KHz baud rate.



2. Host computer settings of CANPro protocol analysis platform

First, connect the device and turn on the power.

(1) Start the device and set the same value of baud rate with host computer RS485_BaudRate. If the RS485_BaudRate is set to 2, please set baud rate to 500K KHz. Other options are default. There is heartbeat message feedback.

(2) The frame ID is hexadecimal. If the ID is 10 (decimal), please input 00 00 00 0A.

(3) Send the command, through CAN bus, the host computer will receive a feedback.

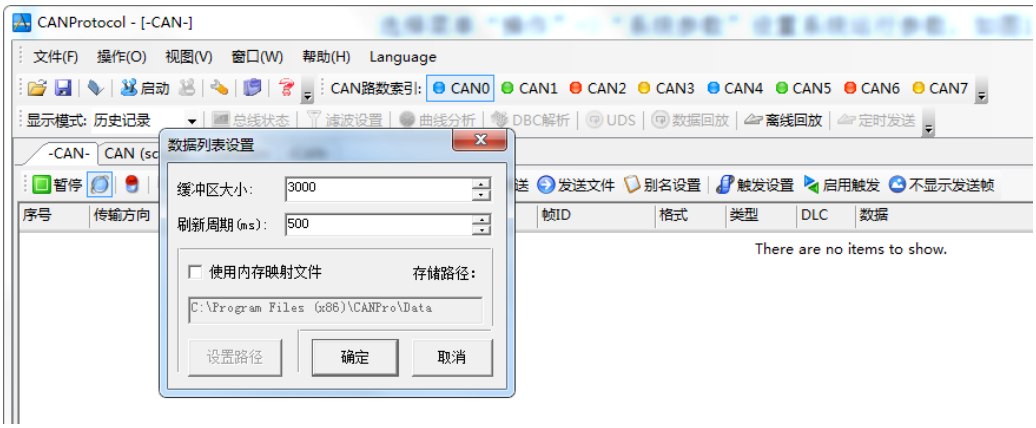
Eg: Send: 00 DA 00 19 00 00 00 3F

Return: 00 DB 00 19 00 00 00 3F

That means the host and the slave can communicate normally.

①Set system parameters

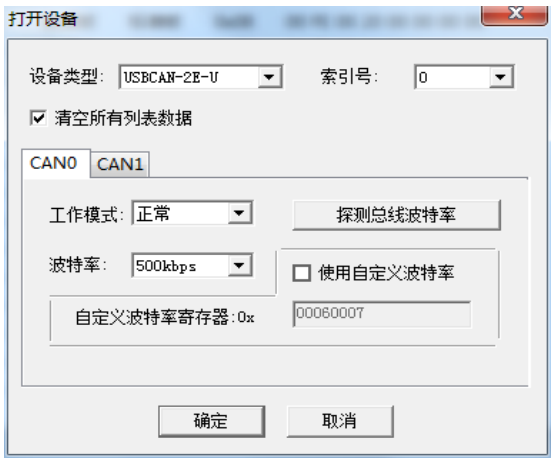
Select menu "Operation" -> "Data List Settings" to set the data list parameters, as shown below:



Buffer size indicates the number of data frames that can be displayed simultaneously in the data list. Refresh period indicates how often the data list is refreshed. The refresh period is preferably set to 200 milliseconds or more, and the recommended value is 500 milliseconds.

②Starting system

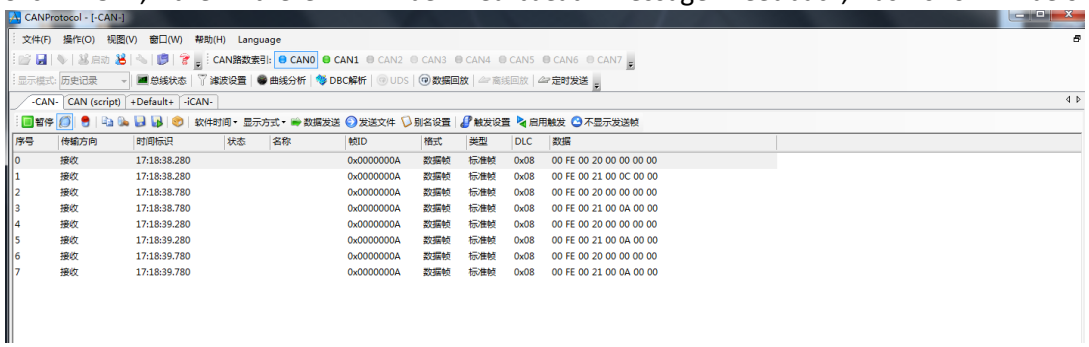
After setting the parameters, you can start the system. Select menu "Operation" -> "Start System", the following picture will pop up:



Select the type of device you want to operate, and select “USBCAN-2E-U” here. When there are several CAN analyzers with same type inserted in computer, for example, there are two CAN analyzers, index numbers 0 and 1 respectively represent the two cards. When there is only one CAN analyzer, the index number can only select 0.

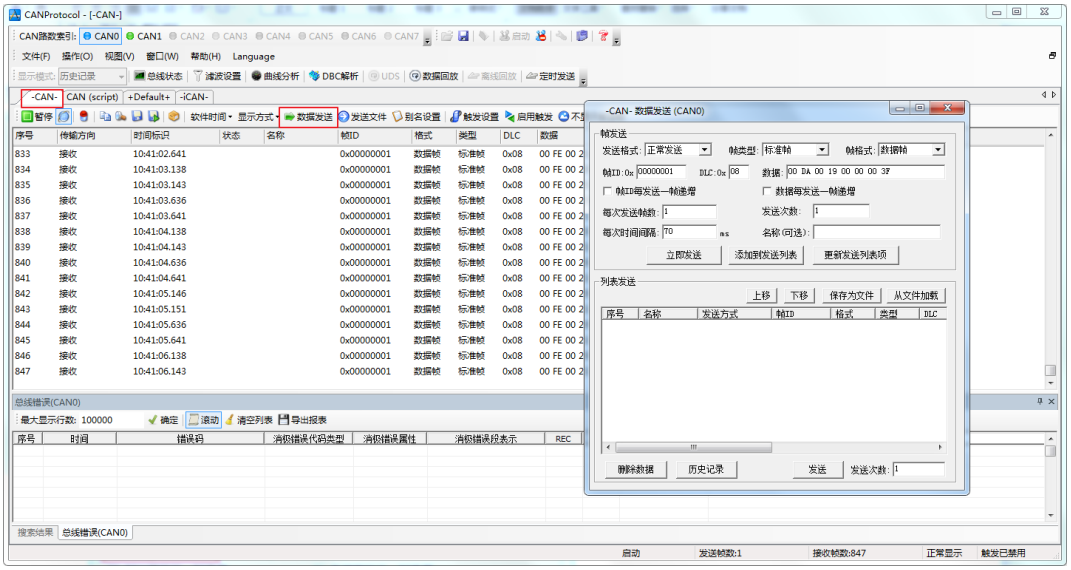
Set the same value of baud rate with host computer RS485_BaudRate. If the RS485_BaudRate is set to 2, please set baud rate to 500K KHz. Other options are default.

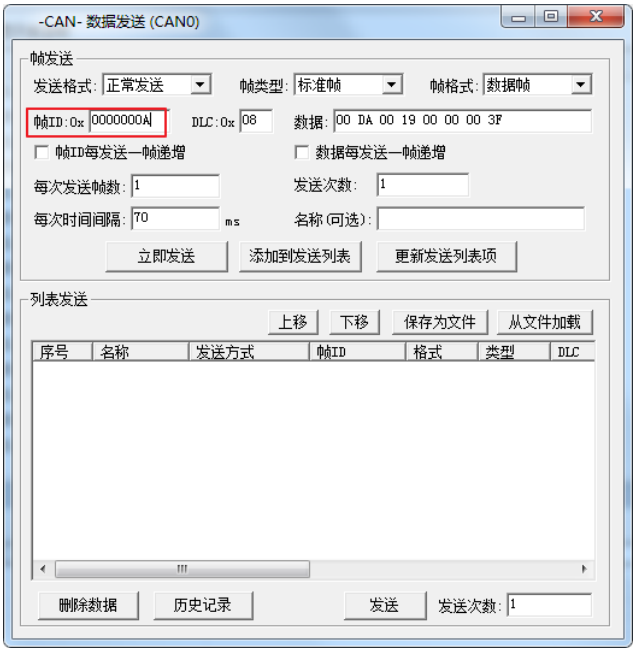
Click "OK", then there will be heartbeat message feedback, as shown below:



③Send data

If there is heartbeat message feedback, select “Data Send” in the CAN window, then “-CAN-Data Send” window will pop up as shown below:



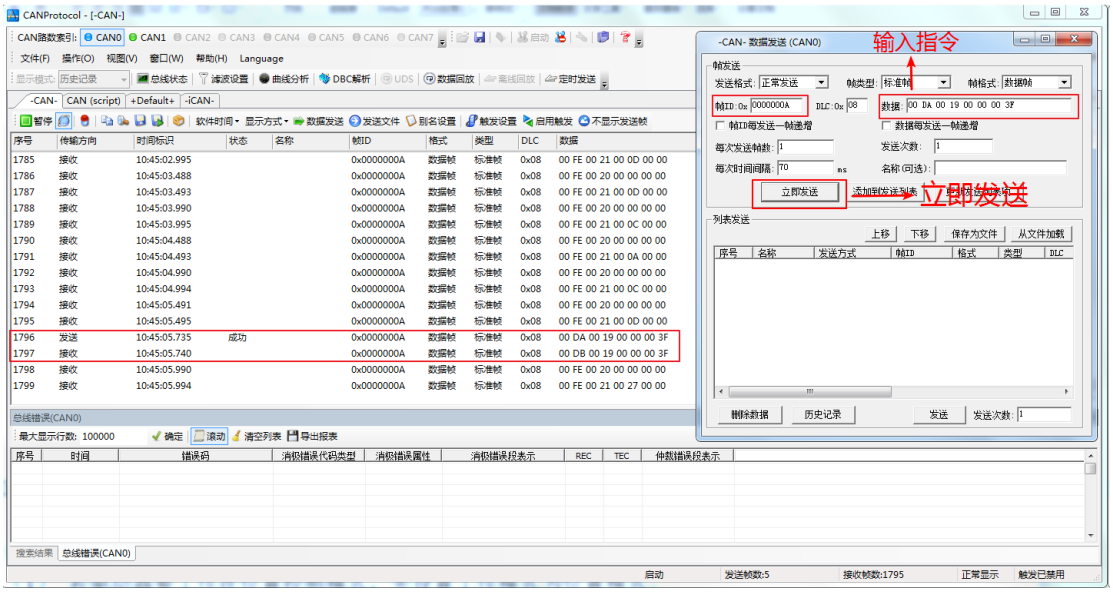


The frame ID is hexadecimal. For example, if the ID is 10 (decimal), then the frame ID input is 00 00 00 0A, and other parameters are default. Write and send command in the data window, through the CAN bus, canpro will receive a feedback.

Eg: Send: 00 DA 00 19 00 00 00 3F

Return: 00 DB 00 19 00 00 00 3F

That means the host and the slave can communicate normally. As shown below:



3. Position mode

Note: for each command sent, you need to check if Canpro received the data. Please send next command after receiving the data.

(1) If the driver need to operate in position control mode, set the working mode to position mode first.

Send 00 DA 00 19 00 00 00 3F, Return 00 DB 00 19 00 00 00 3F.

(2)**Set the ACC/DEC time of the motor.**

Send 00 DA 00 12 00 00 0A 0A, that is 10*100ms, Return 00 DB 00 12 00 00 0A 0A.

(3) **Set the max speed or uniform speed** that the motor is allowed to run. The maximum speed setting is limited to 1000 RPM.

Eg: Send 00 DA 00 14 00 00 01 11, that is 100RPM, Return 00 DB 00 14 00 00 01 11.

(4) **Set the position mode control word**.

Eg: Send 00 DA 00 17 00 00 00 5F. Set the driver to a relative position, Return 00 DB 00 17 00 00 00 5F.

(5) **Enable the motor.**

Send 00 DA 0 0 10 00 00 00 1F, start immediately, the motor is in the lock axis state. Return 00 DB 00 10 00 00 00 1F.

(6) **Send a position command.**

Eg: Send 00 DA 00 16 00 01 E2 40 which the command already contains the direction symbol. That is, the sign of the data following the address 0x0016 is the forward and reverse of the motor position command. Return 00 DB 00 16 00 01 E2 40.

Note: Set the position mode to the opposite direction, set the target position to a negative number, and use the complement of the target position (invert +1).

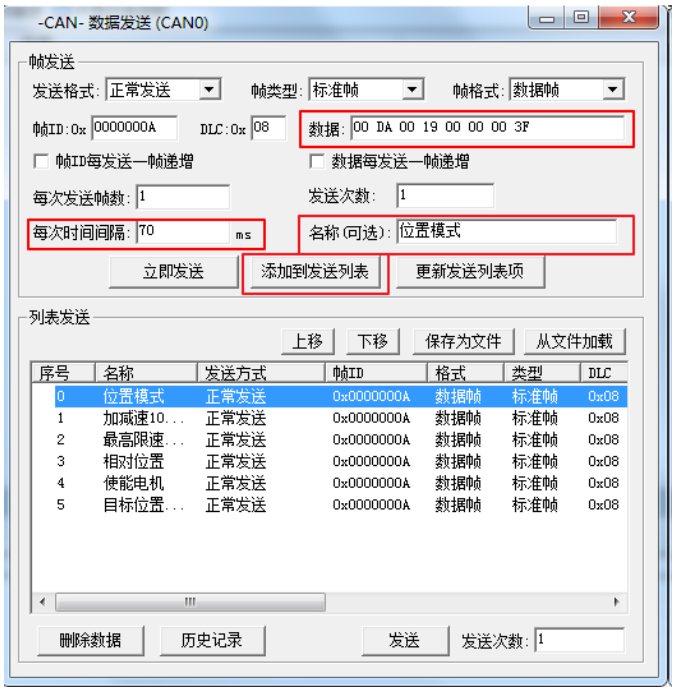
Eg: 00 01 E2 40（hex）= 123456(decimal）

Inverse +1: FF FE 1D BF+1=FF FE 1D C0 = -123456(decimal)

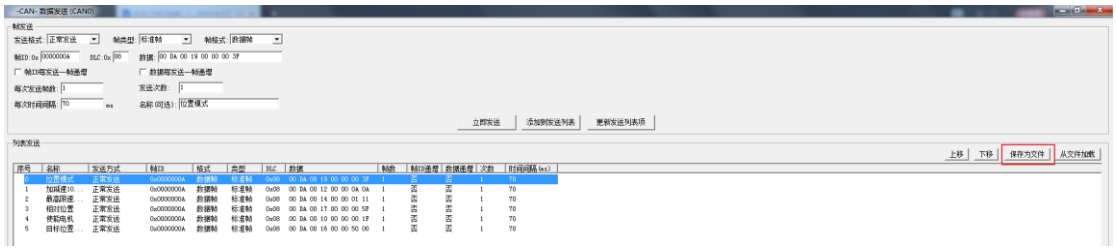
Reverse command sent: 00 DA 00 16 FF FE 1D C0

After the above six steps, the entire process of position control is completed. If you want to

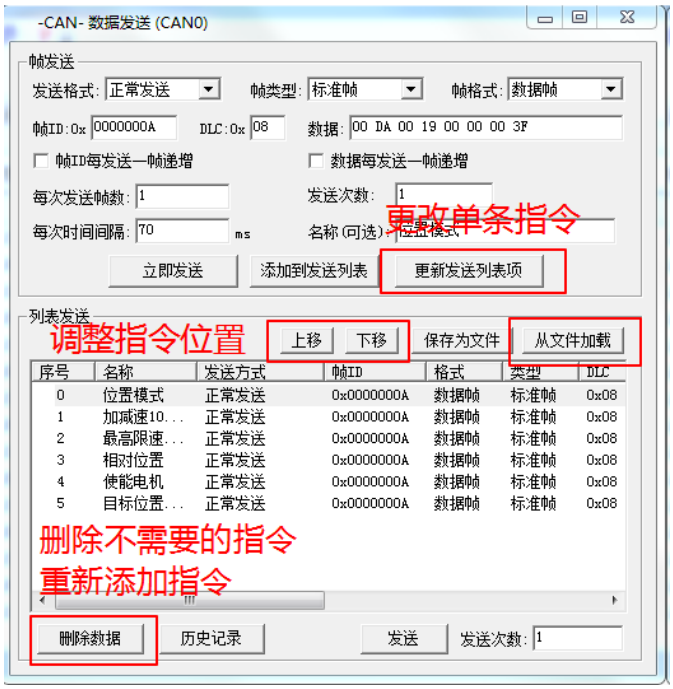
do the next operation, you can repeat the above steps directly. If you do not need to change the motor acceleration and deceleration time, uniform speed, position control word, working mode, directly send the position command again, repeat the sixth step directly. Canpro can save the command, and it can be used by opening it directly from the text next time. Input command will be sent in the data frame, "every time interval" is the time interval of each sending, unit is millisecond, the value is preferably bigger than 70. "Name (optional)" is the name of the command sent. Click "Add to Send List" and the data will be added to the "Send List". The following takes the position mode command as an example, as shown below:



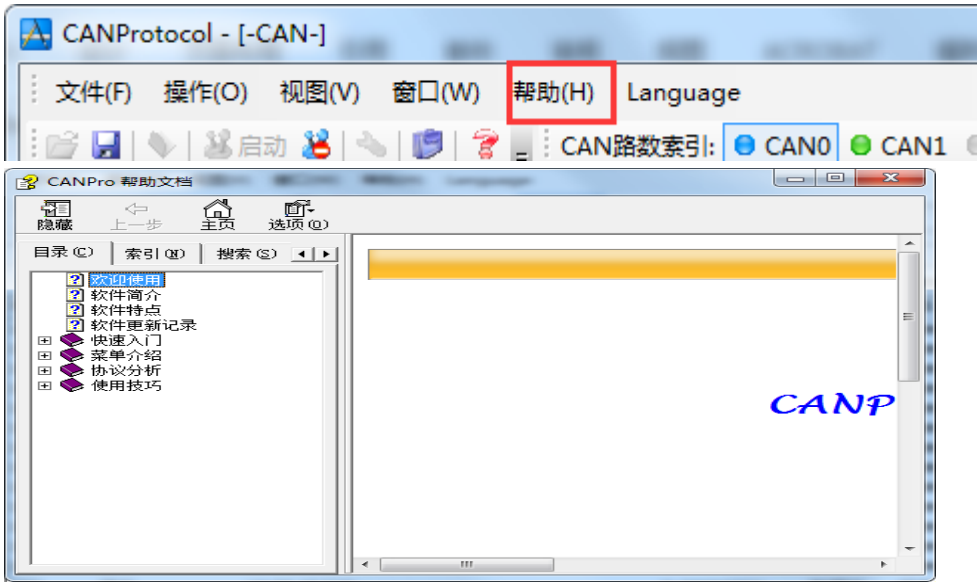
Add the rest of the commands in the position mode to the list. As shown below:



Click "Save File" to save the command of position mode . Then you use click "Load from File" to find the save command file and you can use it directly next time. As shown below:



To understand Canpro better, click "Help" on the menu bar will pop up a help document. As shown below:



4. Speed mode

(1) If the driver need to operate in speed control mode, set the working mode to speed mode first.

Send 00 DA 00 19 00 00 00 2F, Return 00 DB 00 19 00 00 00 2F.

(2) **Set the acceleration/deceleration time of the motor.**

Eg: Send 00 DA 00 13 00 00 0A 0B, the acceleration time is 10*100ms，and the deceleration time is 11*100ms, Return 00 DB 00 13 00 00 0A 0B.

(3) **Set the target speed.** The maximum target speed is 1000 RPM.

Eg: Send 00 DA 00 11 00 00 05 55, Return 00 DB 00 11 00 00 05 55.

(4) **Enable the motor.**

Send 00 DA 00 10 00 00 00 1F, start immediately, the motor is in the lock axis state. If the driver has preset the speed command, the motor will start at the preset speed immediately after the motor is enabled. Return 00 DB 00 10 00 00 00 1F.

(5) Stop the motor.

Send 00 DA 00 10 00 00 00 0F, the motor stops immediately and the motor is offline. Receive 00 DB 00 10 00 00 00 0F. Or send 00 DA 00 30 00 00 00 1F, the motor stops immediately, the motor is not offline and in the lock axis state. Return 00 DB 00 30 00 00 00 1F.

After the above five steps, all the settings of the speed mode have been completed. If you want to change the speed of the motor running in the middle, simply change the target speed. To change other settings, set them as appropriate.

Note:

(1) Driver internal speed conversion: actual speed = set value / 8192 * 3000RPM

(2) Set the motor to reverse and set the speed to a negative number.

Eg: Target speed is 1000RPM, the setting value is 2730 (decimal) = 00000AAA (hexadecimal)

Target speed is -1000RPM and the setting value is FFFFF556.

(3) The maximum speed limit for the hub motor is 1000RPM.

(4) Set the feed forward gain to 0 or 1 according to requirements.

If the feed forward gain is 0, the communication is disconnected, and the motor rotates as usual in the speed mode.

The feed forward gain is 1, the communication is disconnected, the motor will lock the axis, and the speed must be sent every 500ms under normal working conditions, otherwise the motor will also lock the axis.

5. Modify rated current of the driver online

For example, set current 10A, that is, 10000mA. Send 00 DA 00 2d 00 00 27 10.

6. Torque mode

(1) If the driver needs to work in torque mode, set the working mode to torque mode first.

Send 00 DA 00 19 00 00 00 8F, Return 00 DB 00 19 00 00 00 8F.

(2) Set the torque value.

Eg: If setting 2500, which is 8A. Send 00 DA 00 08 00 00 09 C4, Return 00 DB 00 08 00 00 07 C4.

(3) Start the motor, the motor will rotate at the highest speed set inside the driver.

Send 00 DA 00 10 00 00 00 1F, Return 00 DB 00 10 00 00 00 1F.

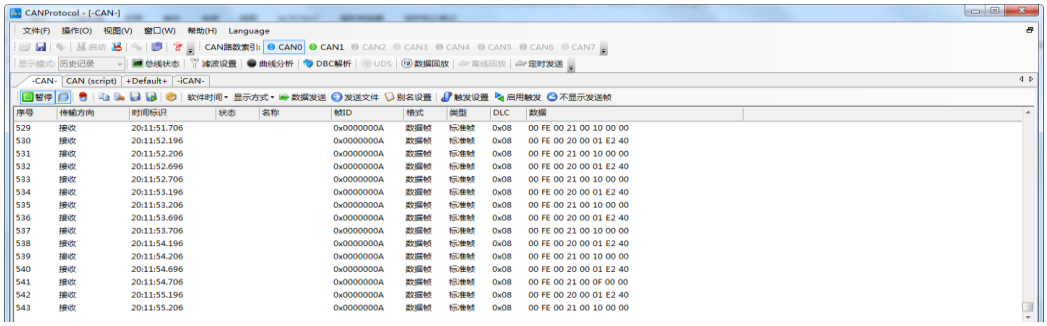
(4) Stop the motor and the motor is offline.

Send 00 DA 00 10 00 00 00 0F, Return 00 DB 00 10 00 00 00 0F. Or send00 DA 00 30 00 00 00 1F, the motor stops immediately, the motor is not offline and in the lock axis state. Return 00 DB 00 30 00 00 00 1F.

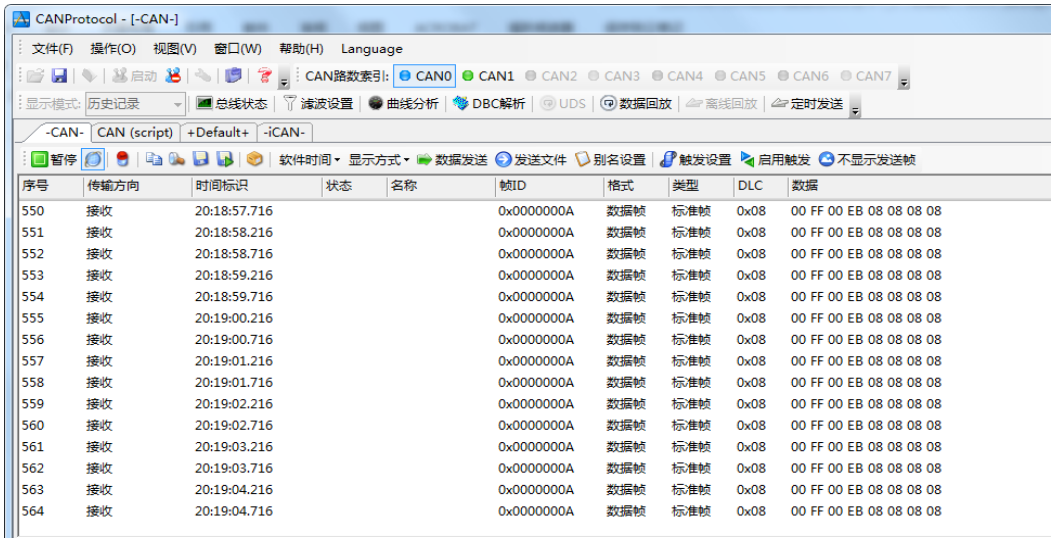
7. Heatbeat message

(1) After completing the setting on the software, if the heartbeat message time is 0, it will not be uploaded. If the heartbeat message time is 500, the driver will upload the heartbeat

message automatically at intervals of 500 milliseconds. The following are heartbeat messages for position, speed, and current.



(2) If the driver alarms, the alarm code will be forced to upload automatically. The following example is an encoder failure.



8.2 RS232 Communication Protocol

8.2.1 RS232 communication protocol

Function Description	Data Add. (A1)	High 8 bits (A2)	Low 8 bits (A3)	Data checksum (A1+A2+A3)	Remark
Motor start	0x00	0x00	0x01	0x01	Write motor enable (PC control mode is valid)
Motor stop	0x00	0x00	0x00	0x00	Write motor disable (PC control mode is valid)
Speed mode selection --- PC digital input	0x02	0x00	0xc4	0xc6	Control mode given command source selection

Position mode selection - external pulse input	0x02	0x00	0xc0	0xc2	Control mode given command source selection
Position mode selection - PC digital input	0x02	0x00	0xd0	0xd2	Control mode given command source selection
Torque mode - PC digital input	0x02	0x00	0xc1	0xc3	Control mode given command source selection
Speed proportional gain	0x40	_____	_____	Take lower 8 bits (A1+A2+A3)	It is recommended to modify according to the actual conditions based on the factory default parameters. Under the default parameter setting, if the motor vibrates, please reduce the value of the speed ratio, and increase the speed integral value appropriately when motor is not vibrating. If you reduce speed proportional gain, but the motor still vibrates, then reduce the position proportional gain value. In the case of non-vibration, increase the speed proportional gain value slowly and reduce the speed integral gain value. Under the condition of satisfying the rigidity, increasing the speed differential gain value appropriately can reduce the motor vibration, and increasing the excessive differential gain value will cause high frequency noise.
Speed integral gain	0x41	_____	_____	Take lower 8 bits (A1+A2+A3)	
Speed differential gain	0x42	_____	_____	Take lower 8 bits (A1+A2+A3)	
Position proportional gain	0x1a	_____	_____	Take lower 8 bits (A1+A2+A3)	
Position differential gain	0x1b	_____	_____	Take lower 8 bits (A1+A2+A3)	
Position feed forward gain	0x1c	_____	_____	Take lower 8 bits (A1+A2+A3)	

Speed mode (valid for PC digital input) --- ACC and DEC setting	0x0a	acceleration	deceleration	Take lower 8 bits (A1+A2+A3)	
		_____	_____		
Position debug mode position given high 16 bits----PC---position	0x50	_____	_____	Take lower 8 bits (A1+A2+A3)	The upper 16 bits of the 32-bit data input
Position debug mode position given lower 16 bits ------- PC-------position	0x05	_____	_____	Take lower 8 bits (A1+A2+A3)	The lower 16 bits of the 32-bit data input
Torque debug mode PC input torque	0x08	_____	_____	Take lower 8 bits (A1+A2+A3)	The input torque 2500 actually corresponds to 8A, and can be set to -7500--+7500
Speed debug mode-----PC digital input	0x06	_____	_____	_____	The set digital 16384 corresponds to actual speed 6000RPM
Looking for Z signal mechanical origin	0x53	0x00	0x00	0x53	When running, send this command, the motor will find the Z signal position automatically
Clear fault	0x4a	0x00	0x00	0x4a	Clear current fault
Absolute position/relative position switching control in position mode	0x51	0x00	0x00	0x51	Absolute position
	0x51	0x00	0x01	0x52	Relative position
Speed limit value in position mode(The actual speed at a given position is reached under	0x1d	_____	_____	Take lower 8 bits (A1+A2+A3)	The set digital 16384 corresponds to the actual speed 6000RPM

the position command)					
Modify the rated current	0x2d	_____	_____	Take lower 8 bits (A1+A2+A3)	Set the value unit to mA
Read monitoring parameters	Send 0x80 0x00 0x80, the driver will return the corresponding monitoring information				
Fault status	0x80	0x00	Status_word	Take lower 8 bits (A1+A2+A3)	Status_word is the custom parameter
	Internal Control (PC): Send：80 00 80 Return：80 00 01 81 E1 00 19 FA E2 00 18 FA E4 02 8B 71 E6 00 00 E6 E7 21 8A 92 E8 00 00 E8 E9 1B 24 28 External Control (PLC): Send：80 00 80 Return：80 00 81 01 E1 00 19 FA E2 00 18 FA E4 02 8B 71 E6 00 00 E6 E7 21 8A 92 E8 00 00 E8 E9 1B 24 28				80 00 01 81 80 00 81 01 Motor starting state E1 Bus voltage address E2 Output current address E4 Output speed address E6 Position given high 16-bit address E7 Position given low 16-bit address E8 Position feedback high 16-bit address E9 Position feedback low 16-bit address
	The fault information corresponding to each bit of Status_word is as follows (high effective): Status_ov_i = Status_word^1, over-current Status_ov_u = Status_word^2, over-voltage Status_err_enc = Status_word^3, encoder error Status_ov_q = Status_word^5, under-voltage Status_ov_load = Status_word^6, overload				
Bus voltage	0xe1	_____	_____	Take lower 8 bits (A1+A2+A3)	(tolerance ±2V)
Output current	0xe2	_____	_____	Take lower 8 bits (A1+A2+A3)	The actual current by 100 times reduced
Output rotate speed	0xe4	_____	_____	Take lower 8 bits (A1+A2+A3)	The returned digital 16384 corresponds to the actual speed 6000RPM

Position given high 16 bits	0xe6	_____	_____	Take lower 8 bits (A1+A2+A3)	The position is given as 32-bit data, and the actual value is recombined according to the upper 16 bits and the lower 16 bits.
Position given high 16 bits	0xe7	_____	_____	Take lower 8 bits (A1+A2+A3)	
Position feedback high 16 bits	0xe8	_____	_____	Take lower 8 bits (A1+A2+A3)	The position feedback is 32-bit data, and the actual value is recombined according to the upper 16 bits and the lower 16 bits.
Position feedback low 16 bits	0xe9	_____	_____	Take lower 8 bits (A1+A2+A3)	

Individually readable monitoring parameter

Parameter	Address	Command Format	Remark
Alarm status	0x60	Internal Control (PC): Send 0x60 0x00 0x00 0x60 Return 60 60 80 00 00 80 60 60 80 00 01 81 60 60 80 00 02 82 60 60 80 00 04 84 60 60 80 00 08 88 60 60 80 00 10 10 60 60 80 00 20 A0 60 60 80 00 40 C0 External Control (PLC): Send 0x60 0x00 0x00 0x60 Return 60 60 80 00 80 00 60 60 80 00 81 01 60 60 80 00 82 02 60 60 80 00 84 04 60 60 80 00 88 08 60 60 80 00 90 10 60 60 80 00 A0 20 60 60 80 00 C0 40	Stop state Startup state Overcurrent Overvoltage Encoder error Overheat Undervoltage Overload
Bus voltage	0x61	Send 0x61 0x00 0x00 0x61 Return 61 61 E1 00 19 FA	Voltage = 0x19 corresponds to 25V
Output current	0x62	Send 0x62 0x00 0x00 0x62 Return 62 62 E2 00 0A EC	Current = 0x0A corresponds to 10mA
Motor speed	0x63	Send 0x63 0x00 0x00 0x63 Return 63 63 E4 01 12 F7	Actual speed = set value / 8192 * 3000RPM

			0x01 0x12 corresponds to 274 Speed = 274 / 8192 * 3000 = 100rpm
Position given	0x64	Send 0x64 0x00 0x00 0x64 Return 64 64 E6 00 00 E6 E7 27 10 1E	After E6 is the position high 16 bits 0x00 0x00 After E7, the position is lower 16 bits 0x27 0x10 = 10000 counts
Position feedback	0x65	Send 0x65 0x00 0x00 0x65 Return 65 65 E8 00 00 E8 E9 27 09 19	After E8 is position feedback high 16 bits After E9 is position feedback low 16 bits

Description:

(1) The data command format received by the controller is: address + data 8 bits + data low 8 position + data checksum (take the lower 8 bits of the first 3 data sums), If the host computer send it correctly in this format, the driver immediately returns the two addresses of the command to the host computer, indicating that the driver has successfully received the command. For example: the host computer sends: 0x09 0x32 0x32 0x6d the driver returns to the host: 0x09 0x09 This indicate the driver has finished receiving command.

(2) When selecting the position debug mode, the host computer sends a control command through the serial port. The setting sequence is: set the driver to the position debug mode (send 0x02 0x00 0xd0 0xd2) ---------- set its acceleration and deceleration ---------- motor start (0x00 0x00 0x01 0x01)---------- Position given the upper 16 bits ---------- position given the lower 16 bits.

(3) If all parameters have been set when the driver is debugging parameters, such as the acceleration and deceleration, control mode, speed limit (the factory has defaulted to a suitable value). In this situation, only need to set the motor to start, then send position to the positioning.
Step: Motor start (0x00 0x00 0x01 0x01) ---------- Position given high 16 bits ---------- position given lower 16 bits.

(4) The input length given by the position is a 32-bit data, which needs to be decomposed into high 16 bits and low 16 bits when transmitting. Moreover, the 16-bit data is also decomposed into high-eight and low-eight bits meanwhile. The data splitting is done by the data format generator in the file. As long as the user writes the data to be sent on it, it can be automatically split into the format used for communication.

(5) Use of the data format generator. For example, set the acceleration and deceleration to 50, and enter it in the 16-bit data generation box: Address: 09 Acceleration write 50 Deceleration write 50 then display. At this time, the data format is generated 0x09 0x32 0x32 0x6d and then the host computer sends this command to set the acceleration and

deceleration in the position debug mode.

(6) Set the 32-bit position given command. The 32-bit data corresponds to the pulses number. For example, when the numerator denominator is set to 1 in the set electronic gear. When the encoder lines number is 2500, the pulses number required for one rotation of the motor is 10000. When the position reference is written as 10000, the driver starts at the mechanical zero point and the motor rotates once. When 100000 is written, the motor rotates 10 turns. If 0 is written, the motor rotates to the position just start.

(7) About the position mode, whether the send position command is an absolute position or a relative position. When the send command is 0x52 0x00 0x00 0x52, the send position is absolute position. When the send command is 0x52 0x00 0x01 0x53, the position is relative position.

(8) In position mode, the maximum stable speed limit of the output motor is determined by VLimit. The send command is (0x1d set value high 8 bits set value lower 8 bits checksum), where the limit speed corresponds to the set value = (required set limit speed / 6000) * 16384, the obtained data is rounded off. For example, if the motor require 3000 RPM, the set value is 8192. If 1RPM is required, the set value is 3 (rounded).

(9) About finding the mechanical Z signal origin, after the parameter configuration is completed and after sending the origin operation (0x53 0x00 0x00 0x53), then send the motor start (0x00 0x00 0x01 0x01). The motor will rotate slowly until find the Z signal origin, and then it will not move. The hub motor has no origin signal.

(10) About monitoring command. The command to send a monitor command is (0x80 0x00 0x80). After receiving the command, the driver will return the following data: fault information, bus voltage, output current (amplified by 100 times, the actual display current need to be divided by 100, for example, the receipt is 123, actual current is 1.23A), and the output speed (The output speed is digital, and the conversion relationship is actual speed = (digital speed / 16384) * 6000), the current position given value is high 16 bits, the current position given value is low 16 bits, and the current position feedback value is high 16 bits. The current position feedback value is low16 bits. Please refer to the above table for the corresponding relationship. The format returned is 4 data per frame. The format is: Address Data high 8 Bits Data low 8 bits Checksum (take the low 8 bits).

(11) Parameters that need to take effect after a power failure restart: Control mode conversion (PC/PLC), CAN ID change.

PC control monitoring return information example description:

Internal alarm code Add 00 00 00 00 to alarm code in internal control mode

External control code Add 00 00 80 00 to alarm code in external control mode

Internal control (PC control):

Send: 80 00 80

Return: 80 00 01 81 E1 00 19 FA E2 00 18 FA E4 02 8B 71 E6 00 00 E6 E7 E7 27 10 1E E8 00 00 E8 E9 27 10 20

Stop status: 80 00 00 80 Status_word 00 00 00 00

Startup state: 80 00 01 81 Status_word 00 00 00 01

Overcurrent: 80 00 02 82 Status_word 00 00 00 10
Overvoltage: 80 00 04 84 Status_word 00 00 01 00
Encoder error: 80 00 08 88 Status_word 00 00 10 00
(Position error is too large)
Overheat: 80 00 10 90 Status_word 00 01 00 00
Undervoltage: 80 00 20 A0 Status_word 00 10 00 00
Overload: 80 00 40 C0 Status_word 01 00 00 00

External control (PLC control):

Stop status: 80 00 80 00 Status_word 00 00 00 00
Startup state: 80 00 81 01 Status_word 00 00 00 01
Overcurrent: 80 00 82 02 Status_word 00 00 00 10
Overvoltage: 80 00 84 04 Status_word 00 00 01 00
Encoder error: 80 00 88 08 Status_word 00 00 10 00
(Position error is too large)
Overheat: 80 00 90 10 Status_word 00 01 00 00
Undervoltage: 80 00 A0 20 Status_word 00 10 00 00
Overload: 80 00 C0 40 Status_word 01 00 00 00

(12)About the rated current modification. Send command (0x2d set value high 8 bits set value low 8 bits checksum);
The set value unit is mA, e.g. setting 8000mA, send (0x2d 0x1f 0x40 0x8c).

(13)About torque debug mode - PC input. The set sequence is: set the driver in torque debug mode - PC input
(send 0x02 0x00 0xc1 0xc3)--Set the torque value---Start (0x00 0x00 0x01 0x01)--Stop (0x00 0x00 0x00 0x00)

8.2.2 RS232 Communication application example

The following command data is hexadecimal, and the host computer sets the control mode PC control.

1. Control step in position mode is as follow:

1. Position mode: 02 00 D0 D2
2. Position mode limit speed(speed choose one among three):
   (1)30RPM： 1D 00 52 6F
   (2)60RPM： 1D 00 A4 C1
   (3)120RPM: 1D 01 48 66
   Calculation: set value = 60/6000*16384=164=0XA4
   Checksum: 0x1D+0x00+0xA4 take low 8 bits, and the last command is 1D 00 A4 C1
3. Enable, lock motor shaft: 00 00 01 01
4. Position given

(1)50 00 00 50 position given 10000 high 16 bits
05 27 10 3C position given 10000 low 16 bits
(2)50 FF FF 4E position given -10000 high 16 bits
05 D8 F0 CD position given -10000 low 16 bits
Calculation: 10000=0X00002710 send high 16 bits and low 16 bits separately

high 16-bit command checksum: 0x50+0x00+0x00=0X50
The command sent: 50 00 00 50
low 16-bit command checksum: 0x05+0x27+0x10=0X3C
The command sent: 05 27 10 3C

Negative position calculation:
e.g. -10000=0xFFFFD8F0, i.e. 0x00002710 Inverse and add one.

5. The speed limit can be modified during a given position.

2. Control step in speed mode is as follow:

1. PC speed mode: 02 00 C4 C6
2. Set the acceleration/deceleration time: 0A 14 14 32 The former 0x14 is the acceleration time 20*100mS, and the latter 0x14 is the deceleration time 20*100mS.
3. Set the rotate speed (RPM) reference as follows:

Rotate Speed(RPM)	Command	Remark
50	06 00 88 8E	
-50	06 FF 78 7D	
100	06 01 11 18	
-100	06 FE EF F3	
200	06 02 22 2A	
-200	06 FD DE E1	

Calculation: 100RPM/3000*8192=273, the hexadecimal of 273 is 111.
Data = (actual speed / 3000 * 8192) then transferred to hexadecimal.

4. Enable motor: 00 00 01 01
According to the demand, the feedforward gain of hub motor's position ring will be set to 1 by default. In speed mode, you need to send 80 00 80 in 1 second, otherwise the motor axis will be locked. When the current feed gain is set to 0, the motor axis will not be locked if not sending 80 00 80. (The value of the feed forward gain can be set to 0 or 1)
5. Shutdown motor: 00 00 00 00

3. Example of current modification is as follow:

Current(mA)	Command	Remark
500	2d 01 f4 22	
1000	2d 03 e8 18	
3000	2d 0b b8 f0	
5000	2d 13 88 c8	
8000	2d 1f 40 8c	

4. Torque mode

1. PC torque mode: 02 00 C1 C3
2. Set the torque value as follow:

Set value	Corresponding Current (A)	Command	Remark
0	0	08 00 00 08	
200	0.64	08 00 C8 D0	
500	1.6	08 01 F4 FD	
1000	3.2	08 03 E8 F3	
2500	8	08 09 C4 D5	

3. Start motor: 00 00 01 01
4. Stop motor: 00 00 00 00

5. Alarm clearing: 4A 00 00 4A

For more detailed technical support needs, please contact the manufacturer's technical support Phone: 13164768688(Mr. Zhang) and Email: sales@zlingkj.com.